Chulalongkorn University · Department of Chemical Engineering

Natural Draft Fired Heater Ver 1 · H-100 Operator Training Simulator

Procurement, completion and commissioning plan

NDFH-V1-CM-PCM-001 · Rev 0 · Issued for teaching

Rev	Date	Description	Originator	Checker	Approver
0	25-Oct-26	Issued for teaching	A. Soottitantawat	—	—

Title block	
Client	Chulalongkorn University · Department of Chemical Engineering
Project	Natural Draft Fired Heater Ver 1 · H-100 Operator Training Simulator
Document no.	NDFH-V1-CM-PCM-001
Title	Procurement, completion and commissioning plan
Drawings	PFD-100 rev 1; P&ID-001 rev C (screen drawing); NDFH-V1-PID-001 rev E (EPC alternative)
Scope	Purchase packages, inspection and tests, mechanical completion, pre-commissioning, dry-out, commissioning, handover.

All numbers in this document come from the Natural Draft Fired Heater Ver 1 simulator model (design case and model runs) or are stated as assumed. It is a teaching document, not a design for construction.

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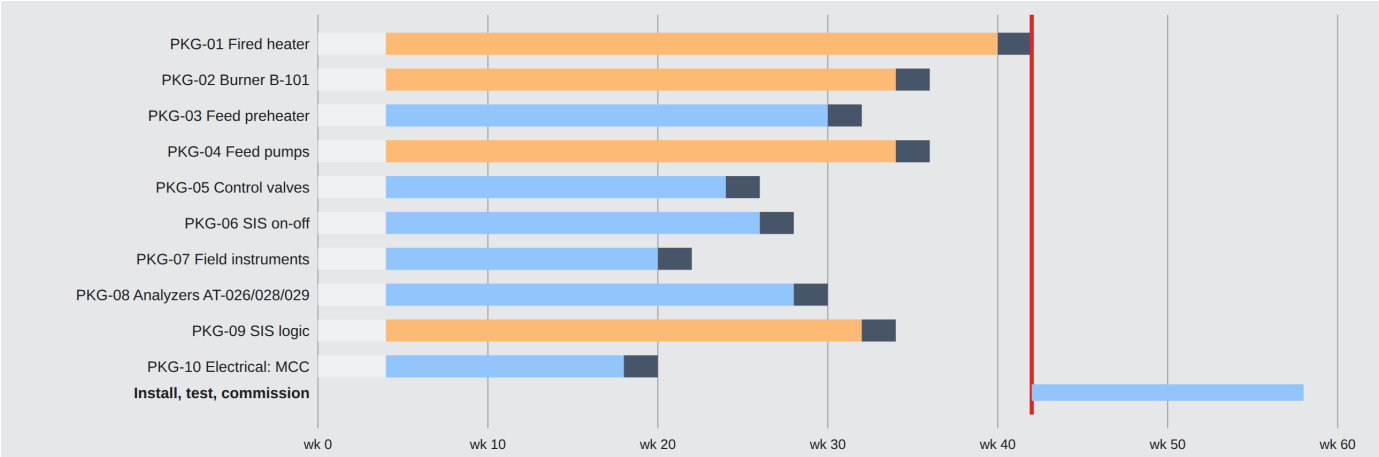
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1 · Purchase packages

Package	Scope	Specification	Lead time wk (A11)	Long lead
PKG-01	Fired heater H-100 (coil, casing, refractory, stack, damper)	ME-DS-101, API 560	36	yes
PKG-02	Burner B-101 and BMS with flame scanner	ME-DS-102, NFPA 85/86	30	yes
PKG-03	Feed preheater E-100	ME-DS-103, TEMA	26	no
PKG-04	Feed pumps P-100A/B with motors	ME-DS-104, API 610	30	yes
PKG-05	Control valves FV-010, FV-200, FV-300, HV-063, HV-400	IN-INL-001	20	no
PKG-06	SIS on-off valves XV-301/302/303 with solenoids	IN-INL-001, IN-CE-001	22	no
PKG-07	Field instruments and transmitters	IN-INL-001	16	no
PKG-08	Analyzers AT-026/028/029	IN-INL-001	24	no
PKG-09	SIS logic solver (SIL 2 certified)	IN-CE-001, SRS	28	yes
PKG-10	Electrical: MCC buckets, cables, heat tracing	EE-LL-001	14	no



Orders placed in week 4 after FEED approval, 16 weeks of installation and testing (the defaults of the tab).

2 · Mechanical completion and pre-commissioning

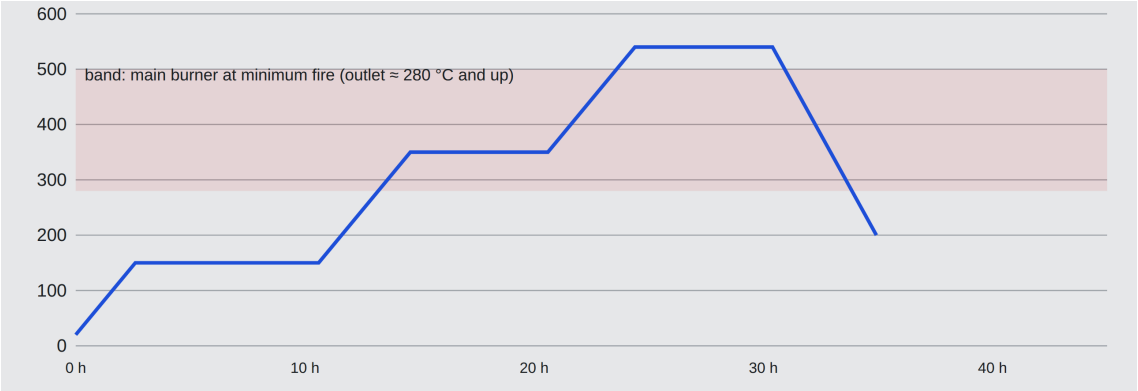
Hydrotest = test factor × design pressure (the stress-ratio correction for hot design is left out).

Item	Design kPa(g)	Factor	Test kPa(g)
Heater coil (API 560)	1700	1.5	2550
E-100 tube side (ASME VIII-1 UG-99)	1700	1.3	2210
E-100 shell side	1000	1.3	1300
Process piping (ASME B31.3)	1700	1.5	2550
Fuel oil piping	2200	1.5	3300

C&E function tests (every trip row of NDFH-V1-IN-CE-001)

Row	Cause	Setting	Expected effect
001	**HS-001** Emergency shutdown push-button (BMS panel / DCS)	Tripped (opened contact)	XV-301/302 close, XV-303 opens, FV-200 and HV-400 open, BMS TRIPPED with first-out
002	**FSLL-008** Feed flow to coil low-low trip (FT-008)	Low-Low 3.4 L/s	XV-301/302 close, XV-303 opens, FV-200 and HV-400 open, BMS TRIPPED with first-out
003	**TSHH-025** Heater outlet high-high trip (TE-025)	High-High 375 °C	XV-301/302 close, XV-303 opens, FV-200 and HV-400 open, BMS TRIPPED with first-out
004	**PSLL-020** Fuel pressure at burner gun low-low trip (PT-020)	Low-Low 40 kPa g	XV-301/302 close, XV-303 opens, FV-200 and HV-400 open, BMS TRIPPED with first-out
005	**PSLL-030** Windbox draft low-low trip (PT-030)	Low-Low 0.5 mmH₂O	XV-301/302 close, XV-303 opens, FV-200 and HV-400 open, BMS TRIPPED with first-out
006	**BSLL-005** Flame failure (scanner BS-005) while firing	Flame lost No flame	XV-301/302 close, XV-303 opens, FV-200 and HV-400 open, BMS TRIPPED with first-out
007	**BSLL-005** Flame not proven at end of trial for ignition	Not proven 10 s trial s	XV-301/302 close, XV-303 opens, FV-200 and HV-400 open, BMS TRIPPED with first-out

3 · Refractory dry-out



A typical vendor curve: 50 °C/h to 150 °C, hold 8 h; to 350 °C, hold 6 h; to 540 °C, hold 6 h; cool at 75 °C/h. The model shows that at minimum fire with feed flowing the main burner already gives an outlet of about 280 °C, so the early holds normally need a temporary burner.

4 · Commissioning and start-up

Step	Practised on the simulator	SOP
Utilities in: instrument air, power, UPS, steam to HV-221 header	—	SOP-H100-01 Pre-startup checks
Loop checks and C&E function test complete; bypasses removed	—	SOP-H100-01 Pre-startup checks
Refractory dry-out (temporary burner or vendor procedure)	—	Vendor procedure
Feed circulation through the coil, FIC-003 above the FSLL setting	Cold startup through the BMS	SOP-H100-02 Cold startup through the BMS
Purge, light-off at low fire, warm-up at a controlled rate	Cold startup through the BMS	SOP-H100-02 Cold startup through the BMS
Line out at 350 °C, tune and hand over to operations	Optimisation: minimum safe excess air	SOP-H100-07 Excess-air optimisation
First planned shutdown to inspect and re-torque	Normal shutdown	SOP-H100-03 Normal shutdown
Trip and restart drill with operations	Emergency shutdown	SOP-H100-04 Emergency shutdown and trip response; SOP-H100-10 Restart after a trip
Upset drill: recover from a feed push with air leading fuel	Feed-push recovery: safe-park, air leads fuel	SOP-H100-05 Feed rate change

5 · Handover

Stage	What is handed over	Punch-list rule
Mechanical completion	System built to the drawings, tested, clean	Category A closed; B listed
Ready for start-up	Pre-commissioning done, PSSR signed, SOPs and training done	All A closed; trips tested; no bypass
Performance test	Design case held 72 h	Measured against the HMB design case
Final handover	As-built documents, spares, maintenance plan	B items with owner and date

6 · Checks

Check	This document	Compared with	Result
Every trip has a C&E function test	7 trip rows tested	5 trips in the simulator + ESD	✓
Every simulator procedure has a commissioning step	6 steps with a procedure	5 procedures	✓
Coil hydrotest ≥ 1.5 × design (API 560)	2550 kPa(g)	2550 kPa(g)	✓
BMS purge gives ≥ 4 volume changes of the firebox (NFPA 85/86 practice)	300 s at 1700 kg/h purge air = 4.7 changes	firebox 24.9 m³ (Design tab)	✓
SOP register covers start-up, shutdown and trip response	12 SOPs	SOP tab	✓